

Composing the parallel dimensions (concrete)

one PP stage

a DP × TP × EP mesh of GPUs (pipeline divides the model into stages)

stage boundary → next stage (P2P activations)

W (weight matrix) — TP-sharded across the 2 TP rows

GPU 1

(DP rank 1 · TP rank 1)

GPU 2

(DP rank 2 · TP rank 1)

GPU 3

(DP rank 1 · TP rank 2)

GPU 4

(DP rank 2 · TP rank 2)

data batch — DP-sharded across the 2 DP columns

EP: MoE experts sharded across all 4 GPUs — token → expert = all-to-all

experts 1-15 · each GPU holds its slice; attention/shared weights replicated

dimension

PARTITIONED

REPLICATED

COMMUNICATION



TP

PP

DP

EP

CP (opt)

W rows

layers

data batch

MoE experts

token sequence

activations

boundary activations

model + optimizer

attn + weights

model weights

all-reduce

P2P

all-reduce

all-to-all

ring